

BRAC UNIVERSITY
Department of Computer Science and Engineering

Set:A

Examination: Mid Term
Duration: 1 Hour 30 Minutes

Semester: Summer 2026
Full Marks: 25

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin indicate marks. Understanding the question is part of the examination.

Name:	ID:	Section:
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1.

- a. Two laptops, Titan and Falcon, run the same ISA. Titan has a clock rate of 2.8 GHz and CPIs for instruction classes A, B, C, D are 7, 4, 2, 3. Falcon has a clock rate of 3.5 GHz and CPIs for instruction classes A, B, C, D are 5, 3, 2, 2.

A program with 3.0×10^6 instructions is made up of:

- 50% class A
- 20% class B
- 20% class C
- 10% class D

i. Calculate the average CPI for both Titan and Falcon. [1+1]

ii. Calculate the execution time for both systems. [2+2]

iii. If a reference machine takes 120 ms to run the same program, calculate the SPECRatio for Falcon. [2]

- b. A company stores all of its critical customer data on a single hard drive with no backups. One day the hard drive fails, and all the data is permanently lost, causing major disruption to the business. To prevent this from happening again, the company sets up a new system where the same data is automatically copied and stored on two additional separate hard drives. If one drive fails, the data can still be recovered from the other two. [1+1]

Which of the seven great ideas of Computer Architecture is this scenario supporting/violating (in its initial setup vs. after the fix)? **Explain** with proper identification and justification.

2. Assume, $Y = 1.1011 \times 2^{-2}$, $X = 1.0000 \times 2^3$

- a. **Determine** the value of P if $P = X - Y$ using floating point subtraction. **Show** detailed calculation. [2]

- b. In a custom IEEE754 floating point representation where there are 3 bits assigned for the exponent field, check if the answer from 2.a is going to overflow/underflow. [1.5]

- c. We want to convert Y into a custom IEEE754 floating point representation, what would be the minimum number of bits to represent the sign, exponent and fraction value? [1.5]

3.

- a. Determine whether the following statements are true or false, if the statement is **false**, then write the correct answer. [1+1]
- Using SB-format instructions we can only perform conditional jumps.
 - S-Type instruction format uses one destination register and one source register.
- b. **Explain** how the LUI and JALR instructions work together to construct a 32-bit absolute target address and perform a long jump in RISC-V. [2]
- c. Consider the following partially incomplete RISC-V program. An instruction is missing at address 4056. Their corresponding machine codes are provided. [2]

Address(Decimal)	Instruction	Machine code
4032	addi x7, x0, 100	
4036	andi x9, x18, 60	
4040	bne x9,x7, label2	
4044	ld x5, 0(x16)	
4048	addi x5, x5, -1	
4052	sh x5, 0(x16)	
4056		0xFE92BA4D
4060	lh x5, 2(x16)	
4064	srli x5, x5, 1	
label 2:4068	sh x5, 2(x16)	
4072		

Assume that the missing instruction at address 4056 is a **branch** instruction. Opcode for any branch instruction = 1001101. Funct3 for beq = 010, bne = 001, bge = 000, blt = 011.

Decode the machine code into its corresponding assembly instruction.

Note: Use “label4” as the label for the branch instruction and **write** the final answer in your script.

- d. **Convert** the following high level language into its corresponding assemble language: [4]

<pre>sum = calc(1,10); a[5] = sum*127</pre>	<pre>int calc(int start, int size) { int result = 0; for (int i = start; i < size; i++) { if (a[i] == 0) { result += a[i]; } else { result -= a[i]; } } return result; }</pre>	You must consider the following: • a is a double word array; • Base address of array a is in register x15; • variable sum is associated with register x22; • pass the argument using register x13, x14; • variable result also uses register x22 and variable i uses x24; • return the value using register x10.
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BRAC UNIVERSITY
Department of Computer Science and Engineering

Set:B

Examination: Mid Term
Duration: 1 Hour 30 Minutes

Semester: Summer 2026
Full Marks: 25

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin indicate marks. Understanding the question is part of the examination.

Name:	ID:	Section:
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1.

- a. Two servers, Comet and Vega, run the same ISA. Comet has a clock rate of 3.2 GHz and CPIs for instruction classes A, B, C, D are 6, 5, 3, 2. Vega has a clock rate of 4.0 GHz and CPIs for instruction classes A, B, C, D are 4, 3, 2, 2.

A program with 4.0×10^6 instructions is made up of:

- 30% class A
- 30% class B
- 25% class C
- 15% class D

i. Calculate the average CPI for both Comet and Vega. [1+1]

ii. Calculate the execution time for both systems. [2+2]

iii. If a reference machine takes 100 ms to run the same program, calculate the SPECRatio for Comet. [2]

- b. A processor design team is building a branch-heavy workload accelerator. In the initial design, whenever a conditional branch instruction is encountered, the processor stalls the pipeline and waits until the branch outcome is fully determined before fetching the next instruction. As a result, the processor spends many clock cycles idle, reducing overall performance. To improve efficiency, the team adds branch prediction hardware that guesses the most likely branch direction based on the program's past behaviour and immediately fetches and executes instructions along the predicted path. If the prediction is correct, execution continues without delay. If the prediction is wrong, the incorrectly executed instructions are discarded, and execution restarts from the correct path. [1+1]

Which of the seven great ideas of Computer Architecture is this scenario supporting/violating (in its initial design vs. after the fix)? **Explain** with proper identification and justification.

2. Assume, $X = 1.1011 \times 2^{-2}$, $Y = 1.0000 \times 2^3$

- a. **Determine** the value of P if $P = Y - X$ using floating point addition. **Show** detailed calculation. [2]
- b. In a custom IEEE754 floating point representation where there are 4 bits assigned for the exponent field, check if the answer from 2.a is going to overflow/underflow. [1.5]
- c. We want to convert X into a custom IEEE754 floating point representation, what would be the minimum number of bits to represent the sign, exponent and fraction value? [1.5]

3.

- a. Determine whether the following statements are true or false, if the statement is **false**, then write the correct answer. [1+1]
- SLLI follows the same format as well as all other I format instructions.
 - LUI instruction copies the first 20 bits [19:0], then clears the next 12 bits.
- b. **Explain** how the LUI and JALR instructions work together to construct a 32-bit absolute target address and perform a long jump in RISC-V. [2]
- c. Consider the following partially incomplete RISC-V program. An instruction is missing at address 4056. Their corresponding machine codes are provided. [2]

Address(Decimal)	Instruction	Machine code
4032	addi x7, x0, 100	
4036	andi x9, x18, 60	
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4044	ld x5, 0(x16)	
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4060	lh x5, 2(x16)	
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label 2:4068	sh x5, 2(x16)	
4072		

Assume that the missing instruction at address 4056 is a **branch** instruction. Opcode for any branch instruction = 1001101. Funct3 for beq = 001, bne = 010, bge = 011, blt = 000.

Decode the machine code into its corresponding assembly instruction.

Note: Use “label4” as the label for the branch instruction and **write** the final answer in your script.

- d. **Convert** the following high level language into its corresponding assemble language: [4]

<pre>temp = calc(1, 10); sum = temp*258</pre>	<pre>int calc(int start, int size) { int result = 0; for (int i = start; i < size; i++) { if (a[i] == 0) { result += a[i]; } else { result -= a[i]; } } return result; }</pre>	You must consider the following: • a is a double word array; • Base address of array a is in register x15; • Variables sum, temp are associated with register x22, x23; • pass the arguments using register x13, x14; • variable result also uses register x22 and variable i uses x24; • return the value using register x10.
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